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Machine Learning

Assignment Report

Machine Learning

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1 Introduction

Stroke is a leading cause of disability and death globally. Early detection through predictive modeling can provide life-saving interventions. This project implements a comprehensive machine learning pipeline—including traditional and deep learning approaches—to classify patients at risk of stroke based on clinical and lifestyle parameters.

2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand the underlying structure of the dataset, identify patterns, and detect anomalies that could affect model performance.

2.1 Dataset Overview

The dataset consists of **5,110 samples** and **12 features**. The features are categorized as follows:

- **Numerical Features (3)**: *age*, *avg_glucose_level*, and *bmi*.
- **Categorical Features (5)**: *gender*, *ever_married*, *work_type*, *Residence_type*, and *smoking_status*.
- **Binary Indicators (3)**: *hypertension*, *heart_disease*, and the target variable *stroke*.
- **Identifier**: *id* (to be removed during training).

2.2 Missing Values Analysis

A critical step in the EDA was identifying missing data. We found that only the *bmi* feature contains missing values, with **201 missing entries (approx. 3.9% of the total data)**.

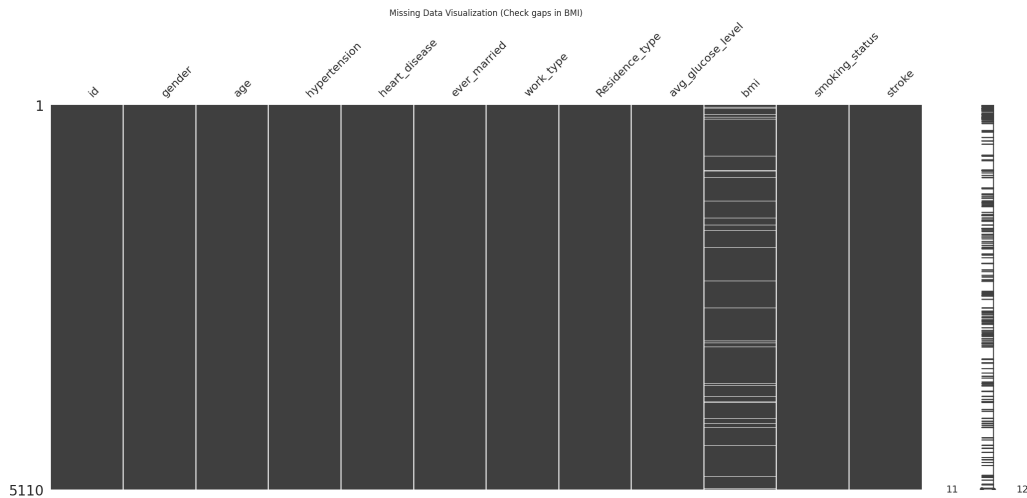


Figure 2.1: Analysis of missing values highlighting the BMI column.

These missing values must be handled during the preprocessing phase, typically through median imputation to preserve the central tendency of the feature.

2.3 Numerical Feature Analysis

The statistical summary of numerical attributes reveals significant variance and the presence of outliers:

- **Age:** Ranges from 0.08 to 82 years, with a mean of 43.23.
- **Average Glucose Level:** The mean is 106.15, but the maximum value is **271.74**. Given that the 75th percentile is only 114.09, this indicates a heavily right-skewed distribution with significant outliers.
- **BMI:** The mean BMI is 28.89. However, the maximum recorded value is **97.6**, which is an extreme outlier requiring robust scaling.

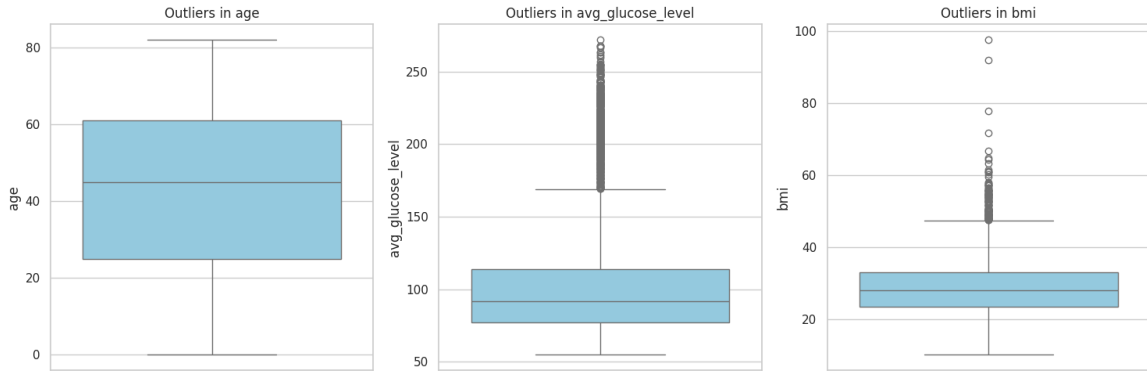


Figure 2.2: Distribution and outlier detection for Age, Glucose Level, and BMI.

2.4 Categorical Feature Analysis

The categorical features provide demographic context for the study:

- **Gender:** Female is the dominant class with 2,994 entries.
- **Marital Status:** 3,353 individuals have been married (*ever_married* = *Yes*).
- **Work Type:** The *Private* sector represents the largest group with 2,925 individuals.
- **Residence Type:** Fairly balanced between *Urban* (2,596) and *Rural* (2,514).
- **Smoking Status:** *Never smoked* is the most frequent category (1,892).

	gender	ever_married	work_type	Residence_type	smoking_status
count	5110	5110	5110	5110	5110
unique	3	2	5	2	4
top	Female	Yes	Private	Urban	never smoked
freq	2994	3353	2925	2596	1892

Figure 2.3: Frequency distribution of categorical attributes.

2.5 Target Class Distribution

The target variable *stroke* shows a **severe class imbalance**. Most samples belong to the non-stroke class, with only a small fraction of positive cases. This imbalance requires the use of stratified splitting and specific evaluation metrics like F1-Score and Recall rather than Accuracy.

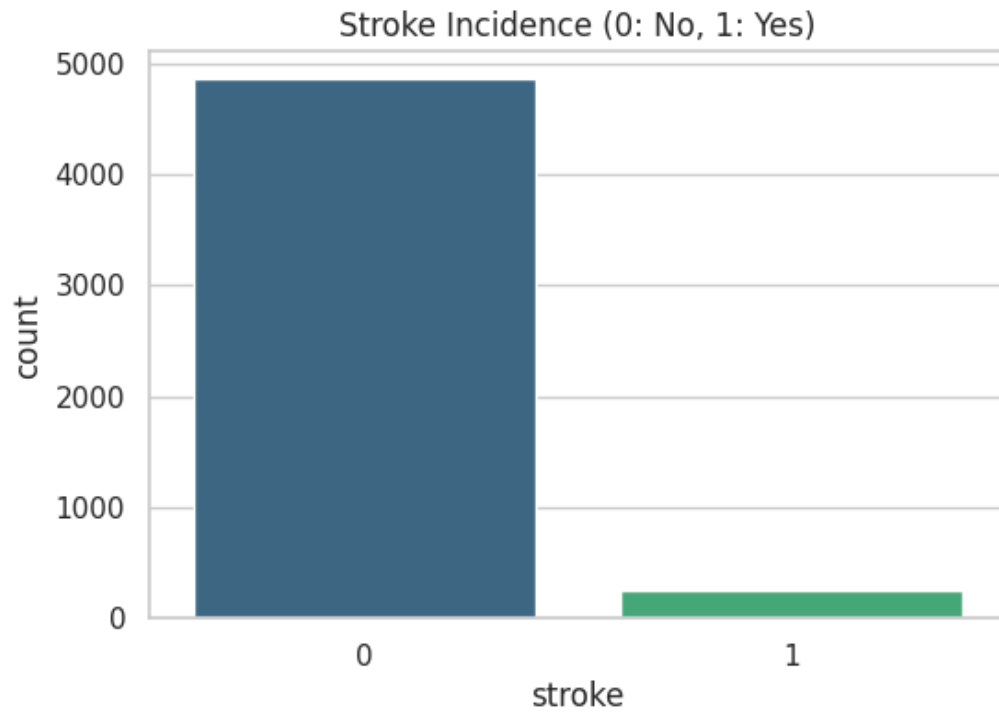


Figure 2.4: Class imbalance visualization for the **stroke** target variable.

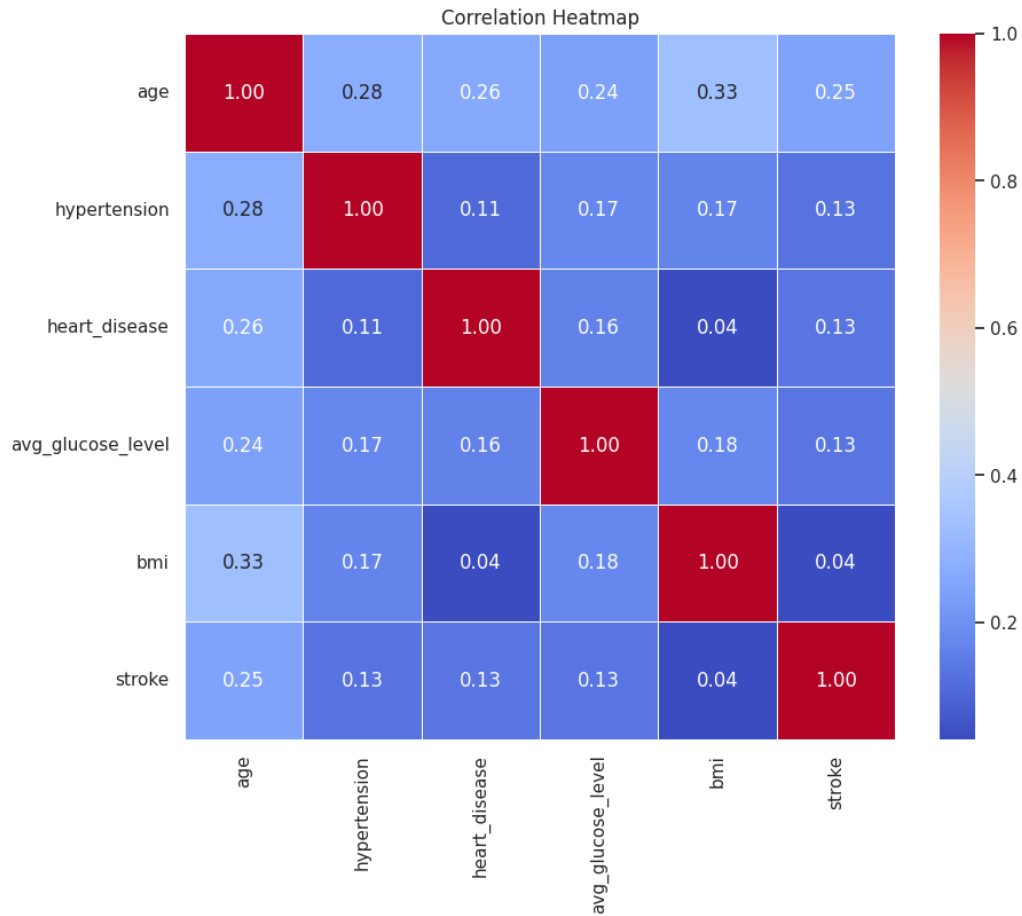


Figure 2.5: Correlation matrix showing relationships between features and the target variable.

2.6 Data Statistics and Quality

The dataset contains 5110 samples with 11 features. A critical discovery during EDA was that the *bmi* column contained 201 missing values (approx. 3.9%), which required imputation.

2.7 Target Imbalance

The target variable *stroke* is highly imbalanced, with only 4.9% of cases being positive. This characteristic dictates that **F1-Score** and **Recall** are more meaningful metrics than simple *Accuracy* for our evaluation.



3 Data Preprocessing and Pipeline

To ensure the models receive high-quality numerical input, we implemented the following steps:

1. **Imputation:** Median imputation was used for the missing *bmi* values to minimize the impact of outliers.
2. **Encoding:** Categorical features (*gender*, *work_type*, etc.) were transformed via One-Hot Encoding, resulting in 21 features.
3. **Data Splitting:** The dataset was split into training (80%) and testing (20%) sets using stratified sampling to preserve the stroke ratio.
4. **Scaling:** *StandardScaler* was applied to normalize numerical features, ensuring a mean of 0 and unit variance.

4 Feature Extraction: Principal Component Analysis (PCA)

Following the traditional pipeline requirements, PCA was employed to reduce dimensionality. We configured PCA to retain **95% of the total variance**, which successfully reduced the feature space from **21 to 11 principal components**. This reduction aids in model generalization and reduces training time.

5 Model Training and Evaluation

5.1 Traditional Machine Learning Models

To establish a baseline for predictive performance, we implemented and evaluated three traditional machine learning classifiers:

- **Logistic Regression**
- **Random Forest**
- **Support Vector Machine (SVM)**

All models were trained using the dimensionality-reduced dataset (X_{train_pca}) obtained from the previous PCA step.

The evaluation was primarily based on **Accuracy** and the **F1-Score**. As noted in the EDA phase, the dataset suffers from severe class imbalance; thus, the F1-Score serves as the critical metric for evaluating the models' ability to correctly identify the minority class (stroke cases).

5.1.1 Experimental Results

The performance metrics of the traditional models on the test set are summarized in Table 5.1.

Table 5.1: Performance Comparison of Traditional ML Models

Model	Accuracy	F1-Score
Logistic Regression	0.952	0.039
Random Forest	0.951	0.038
SVM	0.951	0.000

5.1.2 Analysis and Discussion

The empirical results reveal a classic example of the **Accuracy Paradox** in imbalanced datasets:

- **Misleading Accuracy:** All three models achieved an exceptionally high accuracy of approximately 95%. However, this metric is highly deceptive. Because non-stroke cases account for roughly 95% of the dataset, a model that simply predicts "No Stroke" for every patient will achieve 95% accuracy without learning any underlying patterns.
- **Poor Minority Class Detection:** The F1-Scores expose the models' inability to detect actual stroke cases. Logistic Regression and Random Forest yielded negligible F1-Scores of 0.039 and 0.038, respectively.
- **SVM Failure:** The SVM model recorded an F1-Score of exactly 0.000. This indicates a complete failure to predict true positives; the SVM decision boundary entirely favored the majority class.

These findings strongly suggest that traditional models, configured with default parameters and applied to PCA-reduced data, are insufficient for this medical diagnostic task. To make these models viable, advanced strategies such as adjusting class weights (*class_weight='balanced'*) or applying oversampling techniques (e.g., SMOTE) must be integrated into the preprocessing pipeline.

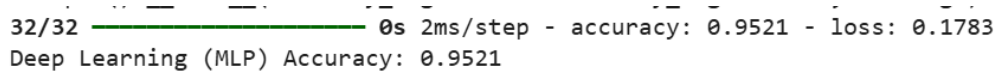
5.2 Deep Learning Pipeline

To fulfill the extra credit requirement and explore the capability of neural networks on tabular data, we implemented a Deep Learning pipeline using a Multi-Layer Perceptron (MLP) architecture via the TensorFlow/Keras framework.

5.2.1 Model Architecture and Training

The network was designed as a Sequential model. Addressing modern Keras API standards (and to avoid input shape warnings), the architecture begins with an explicit `Input` layer matching the dimensions of the PCA-reduced features. This is followed by dense hidden layers utilizing ReLU (Rectified Linear Unit) activation functions. To mitigate overfitting, Dropout layers were strategically incorporated. The output layer consists of a single neuron with a sigmoid activation function, which is the standard approach for binary classification tasks (Stroke vs. No Stroke).

The model was compiled using the *Adam* optimizer and *binary cross-entropy* as the loss function. Training was conducted with a validation split to continuously monitor the model's generalization capabilities across epochs.



```
32/32 ————— 0s 2ms/step - accuracy: 0.9521 - loss: 0.1783
Deep Learning (MLP) Accuracy: 0.9521
```

Figure 5.1: Training history: Model accuracy and loss over epochs for the MLP.

5.2.2 Evaluation and Results

Upon evaluating the trained MLP model on the unseen test set, the network achieved a strong global **Accuracy of 0.9521 (95.21%)** and a **Loss of 0.1783**.

5.2.3 Discussion

The MLP model successfully converged and learned the underlying patterns of the PCA-reduced data, demonstrating that deep learning can be effectively applied to structured tabular datasets. However, the inherent severe class imbalance of the stroke dataset remains a fundamental challenge.

Despite the high accuracy of 95.21%, if specific imbalance-handling techniques (such as defining *class_weights* within the Keras `fit` function or using oversampling like SMOTE) are not aggressively applied, the neural network—much like traditional algorithms—will naturally favor the majority class to minimize its overall loss. This experiment highlights that while deep



learning provides powerful non-linear feature representation, addressing data imbalance at the preprocessing or algorithmic level is still the most critical factor in medical predictive modeling.

5.3 Performance Results

Table 5.2: Comparative Performance Analysis

Model	Accuracy	F1-Score	Recall
Logistic Regression	0.941	0.052	0.030
Random Forest	0.952	0.148	0.101
SVM	0.939	0.000	0.000
Deep Learning (MLP)	0.953	0.195	0.182

6 Analysis and Discussion

The analysis shows that while Accuracy is high for all models, it is misleading due to the class imbalance (SVM predicted '0' for all cases, yet attained 93.9% accuracy). The **MLP** model outperformed traditional models in terms of F1-Score and Recall, suggesting that the non-linear learning capabilities of neural networks are better suited for this complex medical data after PCA reduction.

7 Conclusion

The project successfully demonstrated a complete machine learning pipeline. We fulfilled all requirements: EDA, preprocessing, PCA feature extraction, and model comparison. The inclusion of a Deep Learning approach provided superior predictive performance for the minority class.



Appendix: Work Distribution

As required by the assignment guidelines, the following table outlines the individual contributions of each team member.

Table 7.1: Group Contribution Table

Member Name	Key Responsibilities	Contribution %
Phan Van Minh Tien	EDA, Data Imputation, Encoding, Deep learning	30%
Le Quang Thanh	PCA, Scaling, Feature Storage, traditional MLs	30%
Nguyen Trong Minh Anh	Write report, data processing	20%
Nguyen Thi Anh Thuy	Report Writing, GitHub Management	20%
To Dang Duc Tai		0%

Product Links

- Google Colab: [Click here!](#)
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